College Board AP® Chemistry

Your notes

Elements & Mixtures

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Your notes

Formula Units

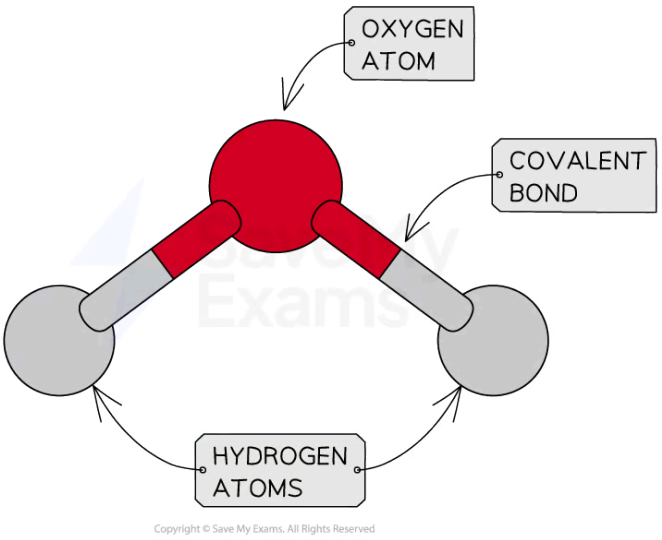
Pure substances and chemical formula

- Atoms are the small building blocks of matter
- One of the broadest classifications of matter are pure substances
- Pure substances can be classified into **elements** or **compounds**
- **Elements** are composed of only one kind of atom
  - Aluminum (Al) or oxygen (O_2)
- **Compounds** have a unique combination of atoms and they can be split into **molecular** or **ionic**
  - This combination of atoms can be represented with a **chemical formula**, in which **atomic symbols** identify the elements in the compound and **subscripts** are used to show how many atoms are present

Molecular compounds

- A molecular compound is composed of individual molecules.
- A **molecule** is formed when two or more atoms are bonded together with a covalent bond
- Water is a molecular compound because its atoms are bonded with covalent bonds
- The chemical formula of water is H_2O

A molecule of water (H2O)



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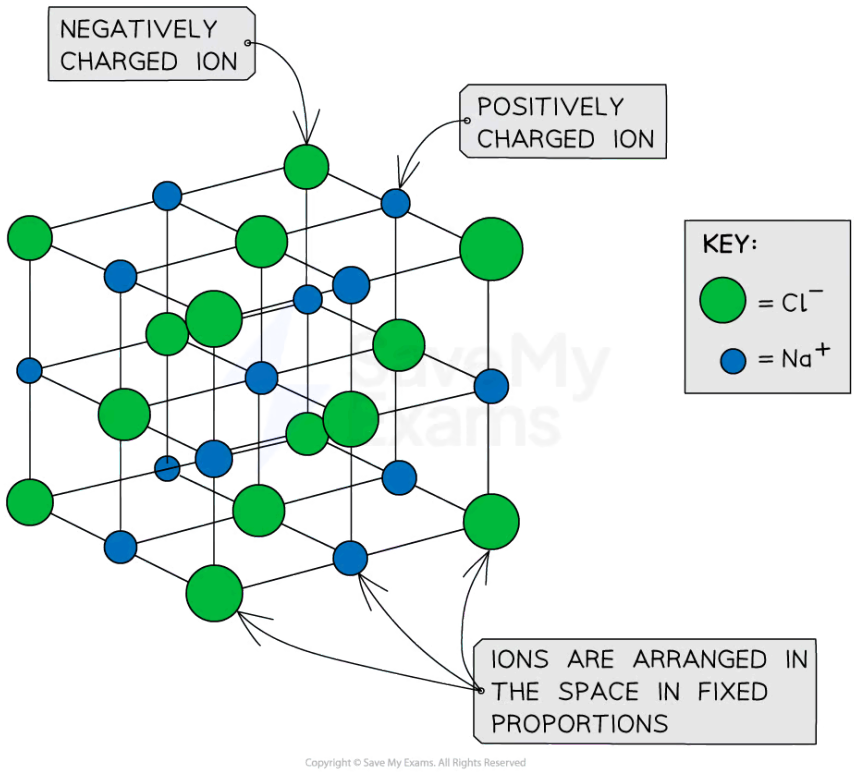
A molecule of water (H_2O) showing 2 hydrogen atoms, 1 oxygen atom and the covalent bonds

Your notes

Ionic compounds

- An ionic compound consists of atoms or ions held together in a formula unit
- A **formula unit** is the arrangement of atoms or ions in fixed proportions by an ionic bond
  - Salt (sodium chloride) is an ionic compound because its ions are held together in a fixed arrangement by ionic bonds
  - The chemical formula of sodium chloride is NaCl

Formula unit of sodium chloride (NaCl)



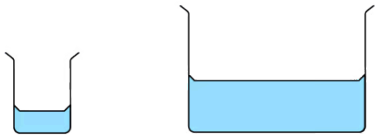
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A formula unit of sodium chloride (NaCl) showing the regular but alternating arrangement of sodium ions and chloride ions

The Law of Definite Proportions

- Elements interact with other elements to form compounds
- The **Law of Definite Proportions** states that the ratio of the masses between the elements in a compound is always the same
- Therefore, a pure compound has the same composition and properties regardless of where it comes from

Law of Definite Proportions



Mass of water (H_2O)	18 g	180 g
Mass of H	2 g	20 g
Mass of O	16 g	160 g
Ratio mass H:mass O	$= \frac{2}{16}$ $= \frac{1}{8}$	$= \frac{20}{160}$ $= \frac{1}{8}$

NO MATTER THE AMOUNT, IF THE PURE SUBSTANCE IS THE SAME, THE RATIO MUST BE EQUAL

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Understanding the Law of Definite Proportions with two samples of H_2O with different masses

Examiner Tips and Tricks

Most of the exam questions regarding the Law of Definite Proportions involve calculating the ratio between the masses of the elements in different compounds. This information can be used to identify if two unknown compounds are the same or not

Empirical Formula

Molecular and empirical formulae

- The chemical formula of a molecular compound is also known as molecular formula
- The **molecular formula** shows the actual number of atoms in a molecule
  - Eg: the molecular formula of butene is C_4H_8
- The **empirical formula** is the simplest formula and shows the simplest ratio of the atoms in a compound
  - Eg: the empirical formula of butene is CH_2
- The chemical formula of an ionic compound is both **empirical** and **molecular formula** at the same time

Examiner Tips and Tricks

The calculation of empirical and molecular formulae are assessed continuously. Expect at least one question related to this topic either in Section I or Section II of your exam

Your notes

Calculating the empirical formula

- The empirical formula shows the smallest ratio between the atoms in molecule or a formula unit
- It can be calculated from the masses of each element in a sample of compound
- If percentages are given instead of masses, assume 100 g of sample and then transform all your % to g

Worked Example

Calculate the empirical formula of a compound which contains 8.0 g of calcium and 14.2 g of chlorine

Answer:

	Calcium	Chlorine
Write down the mass of each element in the sample	8.0	14.2
Calculate the moles using the molar mass using *n = m/M*	$n = \frac{8.0 \text{ g of Ca}}{40.08 \text{ g mol}^{-1} \text{ of Ca}}$ $n = 0.20 \text{ mol of Ca}$	$n = \frac{14.2 \text{ g of Cl}}{35.45 \text{ g mol}^{-1} \text{ of Cl}}$ $n = 0.40 \text{ mol of Cl}$
Divide answers by the smaller value	$= \frac{0.2}{0.2}$ $= 1$	$= \frac{0.4}{0.2}$ $= 2$
Use the answers as subscripts	$CaCl_2$	

Worked Example

Calculate the empirical formula of a compound which contains 86.0% of carbon and 14.0% g of hydrogen

Answer:

	Carbon	Hydrogen
Write down the % of each element in the sample	86.0%	14.0%
Assume 100 g of sample and transform % to g	86.0 g	14.0 g
Calculate the moles using the molar mass using *n = m/M*	n=86.0 g of C12.01 g mol-1 of C n=7.16 mol C	n=14.0 g of H1.008 g mol-1 of H n=13.8 mol H
Divide answers by the smaller value	=7.167.16 = 1	=13.87.16 = 1.93 ≈ 2
Use the answers as subscripts	CH_2	

Your notes

Calculating the molecular formula

- The **molecular formula** is a "factor" of the **empirical formula**
- It can be calculated by dividing the **molar mass of the molecular formula** and the **molar mass of the empirical formula**
- This calculation will result in a **whole number** greater or equal than 1

$$\text{whole number} = \frac{\text{molar mass of the molecular formula}}{\text{molar mass of the empirical formula}}$$

- Finally, all the subscripts from the empirical formula need to be **multiplied** times the whole number, obtaining the molecular formula

Worked Example

The empirical formula of an unknown compound is CH_2O, and the molar mass of the compound is 180 g mol^{-1}. Determine the molecular formula of the unknown compound

Answer:

- **Step 1.** Calculate the molar mass (***M***) of the empirical formula
  - ***M*** of empirical formula = (C x 1) + (H x 2) + (O x 1)
  - ***M*** of empirical formula = (12.01 x 1) + (1.008 x 2) + (16.00 x 1)

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- ***M*** of empirical formula = 30.026 g mol^{-1}

- **Step 2.** Calculate the whole number by dividing ***M*** of the molecular formula by ***M*** of the empirical formula. If the whole number has decimal places, round it to the nearest integer

$$\text{whole number} = \frac{180 \text{ g mol}^{-1}}{30.026 \text{ g mol}^{-1}}$$

$$\text{whole number} = 5.99$$

$$\text{whole number} \approx 6$$

- **Step 3.** Multiply the subscripts of the empirical formula times 6
  - Molecular formula = 6 x CH_2O
  - Molecular formula = $C_6H_{12}O_6$
- The molecular formula of the unknown compound is **$C_6H_{12}O_6$**



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Composition of Mixtures

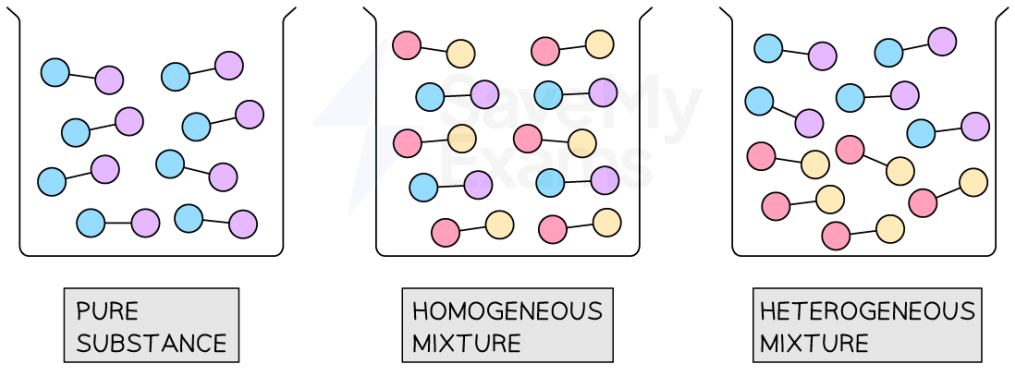
Classification of matter

- Matter is classified in **pure substances** and **mixtures**
- A **pure substance** is matter that has a fixed composition. They contain atoms, molecules or formula units of a single type
  - Aluminum (Al), water (H_2O) or hydrogen gas (H_2), sodium chloride (NaCl) or calcium carbonate ($CaCO_3$) are pure substances
- A **mixture** is matter whose composition is not fixed. Therefore, it contains two or more types of atoms, molecules or formula units
- The molecules or formula units that are part of a mixture are called **components**
  - g. The air we breathe is a mixture in which the components are oxygen, nitrogen and small amounts of other gases
  - g. Water and oil is a mixture between two liquids

Types of Mixtures

- **Homogeneous mixtures** have the same composition throughout
  - The composition is uniform because the components are mixed uniformly in the space
  - g. Air is an homogenous mixture
- **Heterogeneous mixtures** varies from one region of the mixture to another
  - The composition is made from distinct regions because the components separate
  - g. Water and oil is an heterogeneous mixture

Molecules in a pure substance and mixtures



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A pure substance, an homogeneous mixture and a heterogeneous mixture at the molecular level



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Elemental Analysis

Analyzing the composition

- **Elemental analysis** is a technique that can be used to determine the relative composition of elements in a substance and evaluate its purity

Worked Example

A 4.1 g sample of a mixture of calcium chloride ($CaCl_2$) and magnesium fluoride (MgF_2) are found to contain 0.30 g of Ca. What is the percent of $CaCl_2$ in the sample?

Answer:

- **Step 1:** Determine the moles of Ca in the sample
  - n = m / **M**
  - n = 0.30 g / 40.08 g mol^{-1}
  - n = 0.00749 mol of Ca
- **Step 2:** Determine the moles of $CaCl_2$ using the moles of Ca
  - The subscripts show that there is one atom of Ca in one formula unit of $CaCl_2$. Therefore,

$$0.00749 \text{ mol of Ca} \times \frac{1 \text{ mol of } CaCl_2}{1 \text{ mol of Ca}} = 0.00749 \text{ mol of Ca}$$

- **Step 3:** Determine the mass of $CaCl_2$ in the sample by rearranging
  - n = m / **M**
  - m = n x **M**
  - m = 0.749 mol x 110.98 g mol^{-1}
  - m = 0.83 g of $CaCl_2$
- **Step 4:** Calculate the percent of $CaCl_2$ in the sample
  - percent of $CaCl_2$ = (mass of $CaCl_2$ / total mass of sample) x 100
  - percent of $CaCl_2$ = (0.83 g of $CaCl_2$ / 4.1 g of sample) x 100
  - percent of $CaCl_2$ = 20%

Analyzing the purity

- **Purity** is the extent to which a substance contains only a desired chemical compound without any foreign substances
- Purity is often shown as a percentage
- It can be calculated either using the mass (m) of the desired compound or its % composition

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$$\text{Purity} = \frac{\text{m of desired compound}}{\text{total mass of sample}} \times 100$$

$$\text{Purity} = \frac{\text{\% composition of desired compound}}{\text{100 \% of total sample}} \times 100$$

Your notes

Worked Example

5.00 g of a sodium chloride sample were analyzed and it was found that it was contaminated with sulfur. The sample is determined to contain 1.35 g of sodium, 2.00 chlorine, and the rest is sulfur. What is the purity of the sample?

Answer:

- **Step 1:** Identify the desired compound
  - The sample was intended to be sodium chloride so, **NaCl is our desired compound**
- **Step 2:** Calculate the mass of the desired compound
  - The statement lists the masses of sodium and chlorine. Since these are the atoms that are part of NaCl, we add both masses
  - m of desired compound = m of Na + m of Cl
  - m of desired compound = 1.35 g + 2.00 g
  - m of desired compound = 3.35 g
- **Step 3:** Determine the purity using formula that depends on m of desired composition and the total mass of sample

$$\text{Purity} = \frac{\text{m of desired compound}}{\text{total mass of sample}} \times 100$$

$$\text{Purity} = \frac{3.35 \text{ g of NaCl}}{5.00 \text{ g}} \times 100$$

$$\text{Purity} = 67.0\%$$

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